

* Couple the Conf theory to $N=1$ chiral mult. in $D=4$.

$$(Conf) \oplus (N=1 \text{ chiral}) \equiv \text{Compensating Multiplet}$$

Superconf. gauge mult. coupled to Chiral Mult.

$\Downarrow$ Gauge fix

Pure Supergravity

SuperConf. Alg.:

The Superconf. has the conf. algebra $SO(0,2)$ as a subalgebra.

Since the super alg. is generated by matrix elements that are both Bosonic $\oplus$ fermionic, we find, a general matrix is of the form:

$$\begin{pmatrix} \text{Conf. alg.} & Q, S \\ Q, S & R\text{-symm} \end{pmatrix}$$

For $N=2$, $D=4$ the relevant superalgebra is

$$\text{is } \underline{SU(2,2)} \underline{1}$$

main part of

Describes the Bosonic subalgebra $SU(2,2)$

for which $SU(4,2) \simeq SU(2,2)$

Recall: $M^T M = M$ so, in this case

$$U = \text{diag}(-1, 1, \dots, 1, -1)$$

- Refers to the number of susy generators
i.e. $N=1$!

* Bosonic Generators: $N=1$ has an R-symmetry group $U(1)_R$ so the full sym group is

$$(\text{Bosonic subalg.}) = SU(2,2) \times U(1)_R$$

Appears in the block diagonal

* $SU(2,2)$ is the covering space of $SO(4,2)$
i.e. $SO(4,2) \cong SU(2,2)/\mathbb{Z}_2$

As in the simple case of $SO(1,3) \cong SU(2,2)/\mathbb{Z}_2$
this means it contains the same algebra but
allows for fermionic representations
 $\{P_\mu, M_{\mu\nu}, K_\mu, D\} \subset SU(2,2)$

Aside: (R-symmetry)

$$Q_\alpha \mapsto e^{\theta i \gamma_5} Q_\alpha$$

which has the generator T_R . R-sym is
a $U(1)$ chiral symmetry with

$$[T_R, Q] = -i(\gamma_5)_\alpha^\beta Q_\beta \quad (\text{Proof below})$$

$$[T_R, (\text{rest of alg})] = 0$$

So, R-sym is an automorphism of the SUSY

Proof:

$D = \psi: T_A: \text{s.t. } \exists (U_A)_i^j \text{ s.t.}$

$$\begin{aligned} [T_A, Q_{\alpha i}] &= (U_A)_i^j Q_{\alpha j} \\ [T_A, Q_{\alpha}^i] &= (U_A)^i_j Q_{\alpha}^j \end{aligned} \quad \left(\begin{array}{l} \text{Related by charge} \\ \text{conjugate } (C)_{\alpha\beta} \end{array} \right)$$

$$\therefore (U_A)_i^j = -((U_A)^i_j)^*$$

The R-sym generator Q must satisfy the Super-Jacobi identities:

$$[[T_A, T_B], Q] = [T_A, [T_B, Q]] - [T_B, [T_A, Q]]$$

$$\Leftrightarrow f_{AB}^C (U_C) = [U_A, U_B]$$

These are the structure constants we are interested in. Likewise

$$0 = [T_A, \{Q_{\alpha i}, Q_{\beta}^j\}] = \{[T_A, Q_{\alpha i}], Q_{\beta}^j\} + \{[T_A, Q_{\beta}^j], Q_{\alpha i}\}$$

Why does the RHS vanish? By construction

$$[T_A, (\text{Bosonic})] = 0 \text{ and } \{Q, Q\} \sim P_a \text{ so}$$

$$[T_A, P_a] = 0$$

$$\Rightarrow \{(U_A)_i^k Q_{\alpha k}, Q_{\beta}^j\} + \{(U_A)_l^j Q_{\beta}^l, Q_{\alpha i}\} = 0$$

$$\Rightarrow (U_A)_i^k \delta_k^j + (U_A)^j_k \delta_i^k = 0$$

$$\therefore (U_A)_i^j = -((U_A)^j_i)^*$$

i.e. $U_A^\dagger = -U_A$, which are anti-hermitian.

$\rightarrow$ Anti-Hermitian matrices generate $U(N)$

$$(R\text{-symm group}) \cong U(1)$$

We can normalize: $U_1^2 = -U_1^{\dagger 2} = i U$
 generator T_R . Then, given a Majorana
 generator for $N=2$ $Q = P_L Q + P_R Q + Q + Q'$
 $[T_R, Q] = -i P_L Q + i P_R Q = -i \gamma_5 Q \checkmark$

dim	Sector type	R-symm
$D=10$	MW	$SO(N_L) \times SO(N_R)$
$D=9$	M	$SO(N)$
$D=8, 4$	M	$U(N)$
$D=7, 5$	S.M	$USp(N)$
$D=6$	SW	$USp(N_L) \times USp(N_R)$

MW: Maj. Weyl

SM: Symplectic Maj

M: Maj.

SW: Symplectic Weyl

* For $N=1$ chiral: R-symmetry is present
 only when the theory has a Potential
 $W(\Phi, \Phi^\dagger)$ which is of R-charge 2.

Otherwise the R-symmetry is not existent.

→ Here, T_R will act differently w/ the added
 Carl. generators to the Alg.

Why? : Because, to obtain a theory of

Supergravity i.e. gravity coupled

to the Gravitino, we must Gauge the
Symmetries $\Rightarrow T_a$ becomes Gauged
(or LOCAL).

$\rightarrow$ (Gauge T_a at the end of our construction)

The Calm before the storm:

Recall that $ISO(\hat{D-2}, 1)$ has the super-
group extension w/ Generators Q_α .

Now, since we have New generators
 K_a & D must also close the extended
SUSY alg. This is done by introducing
the:

S_α : (Conformal Supercharge)

- S_α 'squares' to a SCT K_a .
- Both S_α and K_a are to be gauged in the
Construction.
- ALL Spinors we consider are Maj.

$$[M_{ab}, Q] = -\frac{1}{2} \gamma_{ab} Q, \quad [M_{ab}, S] = -\frac{1}{2} \gamma_{ab} S$$

$\rightarrow S_\alpha$ are Lorentz spinors.

* Dilations, which define our conf. weight,

give a 5-grading of the Superconf. alg.

h	Gen.
1	P_a
$\frac{1}{2}$	Q
0	D, M_{ab}, T_R
$-\frac{1}{2}$	S
-1	K_a

Not sure, thought -1 must mean $D = -x^a \partial_a$

Since,

$$[D, Q] = \frac{1}{2} Q, \quad [D, S] = -\frac{1}{2} S$$

• For R-symm, we saw that T rotates the susy generators: $[T_R, Q] = -i \gamma_* Q$

Note we used the Normalization $(U_A)' = i$

$\rightarrow$ Here we use $(U_A)' = \frac{3i}{2}$

$$\therefore [T_R, Q] = -\left(\frac{3i}{2}\right) \gamma_* Q$$

$$[T_R, S] = \left(\frac{3i}{2}\right) \gamma_* S$$

$\Rightarrow P_L Q$ and $P_L S$ have opposite $U(1)_R$ charge.

Since K_a lowers the weight by -1 then

$$[K_a, Q] = \gamma_a S$$

$$[P_a, S] = \gamma_a Q$$

• Use $\bar{Q}^\alpha = Q^\alpha = C^{\alpha\beta} Q_\beta$

$$\{Q_\alpha, Q_\beta\} = -\frac{i}{2} (\gamma^a)_{\alpha\beta} P_a$$

$$\{S_\alpha, S^\beta\} = -\frac{i}{2} (\gamma^\alpha)_\alpha^\beta K_\alpha$$

$$\{Q_\alpha, S^\beta\} = -\frac{i}{2} \delta_\alpha^\beta D - \frac{i}{4} (\gamma^{ab})_\alpha^\beta M_{ab} + \frac{i}{2} i (\gamma^\alpha)_\alpha^\beta \bar{T}_R$$

$$\bullet \{[D, S_1], S_2\} + [S_1, S_2], D\} = \{[S_2, D], S_1\}$$

$$-\frac{i}{2} \{S_1, S_2\} = [D, \{S_1, S_2\}] - \frac{i}{2} \{S_2, S_1\} = 0$$

$$[D, \{S_1, S_2\}] = -\{S_1, S_2\}$$

$$\Leftrightarrow \boxed{\{S_1, S_2\} \propto K_\alpha}$$

• We find the factor by looking at other combinations of the Jacobi.

The Storm...

Gauge fields, transformations, & Curvature:

(i) Assign Gauge fields to all generators of the subalgebra.

P_a	M_{ab}	D	K_a	T	Q_α	S
ξ^a	λ^{ab}	λ_D	λ_K^a	λ_T	ϵ_α	$\mathcal{M}$
e_μ^a	ω_μ^{ab}	b_μ	f_μ^a	A_μ	$\psi_{\alpha\mu}$	ϕ_μ

$$\delta = e^A T_A$$

$$\text{and } D_\mu = \partial_\mu - \delta(B_\mu)$$

We split P^a into g_{GFs} and the rest by

$$\delta = e^A T_A = \frac{1}{2} \lambda^{[ab]} M_{ab} + \lambda_D D + \lambda_K^a K_a + \lambda_T T + e^\alpha Q_\alpha + \mathcal{M} S_\alpha$$

Gravitino!

The last two terms are $\bar{E}Q + \bar{M}S$

* For ga's we have $\xi^a = e^a_\mu \xi^\mu$ with

$$\delta(e) B_\mu^A = \partial_\mu E^A + E^C B_\mu^B f_{BC}^A$$

* In the following we underline the terms in which $B_\mu^A \equiv e_\mu^a$, i.e. translations

$$\delta e_\mu^a = -\lambda_b^a e_\mu^b - \lambda_\rho e_\mu^a + \frac{1}{2} \bar{E} \gamma^a \psi_\mu$$

$$\delta \omega_\mu^{ab} = 2 \lambda_c^{[a} \omega_\mu^{b]c} - 4 \lambda_\mu^{[a} e_\mu^{b]} + \frac{1}{2} \bar{E} \gamma^{ab} \phi_\mu + \frac{1}{2} \bar{M} \gamma^{ab} \psi_\mu$$

$$\delta b_\mu = \partial_\mu \lambda_0 + 2 \lambda_\mu^a e_{\mu a} + \frac{1}{2} \bar{E} \phi_\mu - \frac{1}{2} \bar{M} \psi_\mu$$

$$\delta f_\mu^a = -\lambda_b^a f_\mu^b + \partial_\mu \lambda_\mu^a - b_\mu \lambda_\mu^a + \omega_\mu^{ab} \lambda_{ab} + \lambda_0 f_\mu^a + \frac{1}{2} \bar{M} \gamma^a \phi_\mu$$

$$\delta A_\mu = \partial_\mu \lambda_T - \frac{1}{2} i \bar{E} \gamma_* \phi_\mu + \frac{1}{2} i \bar{M} \gamma_* \psi_\mu$$

$$\delta \psi_\mu = -\frac{1}{4} \lambda^{ab} \gamma_{ab} \psi_\mu + \left(\partial_\mu + \frac{1}{2} b_\mu + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab} - \frac{3}{2} i A_\mu \gamma_* \right) E - \gamma_a e_\mu^a \bar{M} - \left(\frac{1}{2} \right) \lambda_0 \psi_\mu + \frac{3}{2} i \lambda_T \gamma_* \psi_\mu$$

$$\delta \phi_\mu = -\frac{1}{4} \lambda^{ab} \gamma_{ab} \phi_\mu + \left(\partial_\mu - \frac{1}{2} b_\mu + \frac{1}{4} \omega_\mu^{ab} \gamma_{ab} + \frac{3}{2} i A_\mu \gamma_* \right) E - \gamma_a f_\mu^a + \frac{1}{2} \lambda_0 \phi_\mu - \frac{3}{2} i \lambda_T \gamma_* \phi_\mu + \lambda_\mu^a \gamma_a \psi_\mu$$

Proof: (i) $\delta b_\mu = \partial_\mu E^0 + E^C B_\mu^B f_{BC}^0$

Note, the only "E⁰" term is given

by $E^0 = \lambda_0$ and λ_T . We only have

$$\partial_\mu \lambda_0 + E^C B_\mu^B f_{BC}^{(0)}$$

so B & C can only be related to objects

$$[1_2, 1_2] \sim \pm i_2 D$$

$$\text{i.e. } [P_0, K_0] = 2(M_{ab} D + M_{ab})$$

$$= f_{P_0 K_0}^{(D)} D$$

$$\partial_\mu \lambda_0 + \lambda_\mu^b e_\mu^a (2M_{ab}) + \dots$$

$$= \partial_\mu \lambda_0 + 2\lambda_\mu^a e_{\mu a} + \dots$$

• The other terms come from

$$\{Q_\alpha, S^\beta\} = -\frac{1}{2} \delta_\alpha^\beta D + \dots$$

$$\therefore \partial_\mu \lambda_0 + 2\lambda_\mu^a e_{\mu a} + \bar{\epsilon} \phi_\mu \left(\frac{1}{2}\right) - \pi \psi_\mu \left(\frac{1}{2}\right)$$

$\rightarrow \pm$ sign occurs from lowering S^β w/
 $C^{\alpha\beta}$. $\square$

(ii) Proof of δe_μ^a

$$\Rightarrow \partial_\mu \xi^a + \partial_\mu \lambda_\mu^a = 0$$

however, in the present case these are
 (?) zero. Not sure why

$\hookrightarrow$ May be from Killing (conf.) eq.

$$\partial_{[a} \xi_{b]} = (M_{ab} \partial_c \xi^c) \frac{1}{D}, \text{ if I had to guess}$$

$$\epsilon_{\beta\mu}^{\alpha} P_\mu^{\beta} + \epsilon_{\beta\mu}^{\alpha} P_\mu^{\beta} f_{\beta c}^{\mu} \neq 0$$

$$P_c: [P_a, M_{bc}] = 2 M_{ab} P_c$$

$$[D, P_c] = P_c, \quad \{Q_\alpha, Q_\beta\} = +\frac{1}{2} (\gamma^a)_{\alpha\beta} P_a$$

$$2\lambda_\mu^a e_{\mu a} + \lambda_\mu^b e_\mu^a + \epsilon^{\alpha\beta} \psi_\mu (\gamma^a)_{\alpha\beta} \left(\frac{1}{2}\right)$$

(iii) b_μ term in $\delta\psi_\mu$

$$w/ [D, Q_\alpha] = \frac{1}{2} \delta_\alpha^B Q_\beta \leftarrow Q_\alpha$$

$$\therefore \delta\psi_\mu^\alpha = b_\mu \epsilon^\alpha + \epsilon^\kappa \beta_\mu^\beta f_{\beta\kappa}^\alpha$$

We want $b_\mu = \beta_\mu^B$ so f_ϵ^α is all

which is left i.e. $\frac{1}{2} \delta_\beta^\alpha = f_\beta^\alpha$

$$\therefore \delta\psi_\mu^\alpha \sim \frac{1}{2} \delta_\beta^\alpha \bar{\epsilon}^B b_\mu = \frac{1}{2} \bar{\epsilon}^\alpha b_\mu \checkmark$$

Where $\epsilon^B = C^{\beta\alpha} \epsilon_\alpha$.

Note the index is up so it really is the transformation for $\delta\bar{\psi}_\mu^\alpha$

Lowering w/ $C_{\alpha\beta}$ gives $\delta\psi_{\mu\alpha}$.

*Recall the Curvature:

$$R_{\mu\nu}^A = 2 \partial_{[\mu} B_{\nu]}^A + g_{\mu\nu}^C B_\mu^B f_{\beta\kappa}^A :$$

$$R_{\mu\nu}(P^a) = 2(\partial_{[\mu} + b_{[\mu}) e_{\nu]}^a + 2 \omega_{[\mu}^{ab} e_{\nu]}^b - \frac{1}{2} \bar{\psi}_\mu \gamma^{ab} f_{\nu}$$

$$R_{\mu\nu}(M^{ab}) = 2 \partial_{[\mu} \omega_{\nu]}^{ab} + 2 \omega_{[\mu}^a \hat{c} \omega_{\nu]}^b + 8 f_{[\mu}^{[a} e_{\nu]}^{b]} - \bar{\psi}_{[\mu} \gamma^{ab} f_{\nu]}$$

...

Constraints:

• We will place constraints to determine the

Gauge field S-susy ψ_μ : The remaining independent fields in the Weyl Mult,

$$e_\mu^a, b_\mu, A_\mu, \psi_\mu$$

$$(i) \quad R_{\mu\nu}(\rho) = 0$$

$$(ii) \quad e_\mu^a \hat{R}_{\mu\nu}(M^{ab}) = i \tilde{R}_\mu^a(T)$$

$$(iii) \quad \gamma^\mu R_{\mu\nu}(Q) = 0$$

→ These constraints are invariant under Q-susy.

• A change in constraint redefines the field that we solve for.

Why the above?

$$\text{For } R_{\mu\nu}(Q) = (\cdot) - 2\gamma_{[\mu}\psi_{\nu]}$$

The γ -trace of $R_{\mu\nu}(Q)$ has the same # of components as the field ψ_μ , which we want to solve for, so $\exists$! a solⁿ of the constraint:

$$(\gamma^\mu)^\alpha_\beta R_{\mu\nu}(Q^\alpha) \sim (\cdot)_{\nu\beta}$$

which is what is meant since $\psi_\mu = (\psi_\mu)_\alpha$

• These constraints generalize the second order formalism in gravity.

Some other formalism: M^{ab} is defined as ρ^{ab}

Second order formulation: ω_μ is defined as a 1-form of e^a_μ i.e. $\omega_\mu^{ab} \in \omega_\mu^{ab}(e)$ so it is not an independent field

Similarly, ω_μ^{ab} , f_μ^a , ϕ_μ are all composite fields i.e. they are f^{ab} of the 'fundamental' gauge fields in the theory: $e^a_\mu, b_\mu, A_\mu, \psi_\mu$

Q: Why do we need these constraints?

→ Before we impose the constraints we have an alg. which is closed on all gauge fields of the Weyl mult. However, translations do not act as they should

• Saw this as $[\delta(\epsilon_1)_{\text{susy}}, \delta(\epsilon_2)_{\text{susy}}] \phi_\mu = 0$

• Gauge translations should be replaced by covariant gct's, which is obtained for the frame field by imposing $R_{\mu\nu}(P^a) = 0$

→ The other fields transform as covariant gct's after all these constraints are imposed.

$R_{\mu\nu}(P^a)$: of SUSY Cont. is similar to $R_{\mu\nu}^P(P^a)$ of the super poincare' except we have an extra b_μ term

• This b_μ can be interpreted as an added term in ω_μ^{ab} i.e. Recall be absorbed this

for pure Conf. gravity:

$$W_{\mu}^{ab}(e, b) = \omega_{\mu}^{ab}(e) + 2 e_{\mu}^{[a} e^{b] \nu} b_{\nu}$$

(i) First Constraint: $R_{\mu\nu}(p^a)$

$$\Rightarrow 2(\partial_{[\mu} + b_{[\mu} \gamma_{\nu]}) e_{\nu]}^a + 2 W_{\mu}^{ab} e_{\nu]} b_b - \frac{1}{2} \bar{\psi}_{\mu} \gamma^a \psi_{\nu} = 0$$

$\Leftrightarrow$

$$W_{\mu}^{ab}(e, b, \psi) = W_{\mu}^{ab}(e, b) + \frac{1}{2} \bar{\psi}_{\mu} \gamma^{[a} \psi^{b]} + \underbrace{\frac{1}{4} \bar{\psi}^a \gamma_{\mu} \psi^b}_{\text{Gravitino torsion}}$$

Gravitino torsion
 T_{μ}^{ab}

• Symbolically rep. the constraint as

$$C(\omega, \psi) = 0 \quad \omega: \text{dependent}, \psi: \text{independent}$$

$\rightarrow$ Assume the constraints are solvable i.e.

$\delta C / \delta \omega$ is invertible.

A solution implies that $\exists$ a $f^{\mu}(\phi)$ $\omega(\phi)$

$$C(\omega(\phi), \phi) = 0 \quad \forall \phi$$

$$\Leftrightarrow \frac{\partial C}{\partial \omega} \frac{\partial \omega}{\partial \phi} + \frac{\partial C}{\partial \phi} = 0$$

Since $\delta \omega = \frac{\partial \omega}{\partial \phi} \delta \phi \Leftrightarrow \delta C \stackrel{!}{=} 0$,

which is equivalent to δ .

(For the spin connection ω_{μ}^{ab} we compute)

δW by using $W(e, b)$ and inserting the transformations of the ind. fields e, b, ψ

* A second way to compute the transformations
 $\rightarrow$ we split the transf. of the dep. fields into the sum of their "old" transformations, dictated by their gauge theory alg. plus extra transf.,

$$\delta W = (\delta_{\text{gauge}})W + \delta_M W$$

where the ladder is inspired by extended cov. der. $\delta(e) B_{\mu\nu}^A = \partial_\mu e^\nu{}^A - \partial_\nu e^\mu{}^A + e^\rho{}^B f_{\rho\mu}^A e^\nu{}^C + e^\mu{}^B M_{\mu\nu}^A$



$$\therefore \delta C = \delta_{\text{gauge}} C + \delta_M C = \delta_{\text{gauge}} C + \frac{\partial C}{\partial W} \delta_M W = 0$$



• First is easy to calc. since the const. is written in terms of curvature:

$$\delta(e) R_{\mu\nu}^A = e^\rho{}^B f_{\rho\mu}^A e^\nu{}^C + e^\mu{}^B M_{\mu\nu}^A$$

and the invertability of $\partial C / \delta W$ allows us then to obtain the new parts of the trans. of the dep. fields

* Let us apply this to the first constraint:

$$(i) R_{\mu\nu}(p^a) = 0$$

$$\text{Since } R_{\mu\nu}(p^a) = e^\rho{}^B f_{\rho\mu}^B e^\nu{}^C + e^\mu{}^B M_{\mu\nu}^B$$

we can have $f_{bc}^a \sim 2\Omega_{ab} [P, M]$

$$[D, P_a] \approx f_{(D)a}^P 1$$

$$\{Q_a, Q^b\} = -\frac{1}{2}(\gamma^a)_\alpha^\beta \quad f_{\alpha\beta}^a = \frac{1}{2}\gamma_{\alpha\beta}^a$$

However cannot be $[P, M]$ or $[D, P_a]$

Since $R_{mn}(P^a) = 0$ thus only $f_{\alpha\beta}^a$ is the only term

$$\therefore \delta_{\text{gauge}} R_{mn}(P^a) = \bar{E}^\alpha R_{mn}(Q_\beta) \left(\frac{1}{2} (\gamma^a)_\alpha^\beta \right)$$

$$\delta_{\text{gauge}} R_{mn}(P^a) = \frac{1}{2} \bar{E}^\alpha \gamma_{\alpha\beta}^a R_{mn}(Q^\beta)$$

From this, we plug into $\star$

$$0 = \frac{1}{2} \bar{E}^\alpha \gamma_{\alpha\beta}^a R_{mn}(Q^\beta) + \left(\frac{\partial R_{mn}(P^a)}{\partial \omega_p^{bc}} \right) \delta \omega_p^{ab}$$

• How do we solve for this? We use the trick: Permute the indices three times and add these cyclically:

ex: $T_{\mu\nu\rho}$:

$$T_{\mu\nu\rho} + T_{\nu\rho\mu} + T_{\rho\mu\nu}$$

$$\begin{aligned} R_{mn}(P^a) &\sim 2\omega_\mu^{ab} e_{\nu d} = \omega_\mu^{ab} e_{\nu d} - \omega_\nu^{ab} e_{\mu d} \\ \frac{\partial R_{mn}(P^a)}{\partial \omega_p^{bc}} &= (\delta_\mu^p \delta_b^a \delta_c^d e_{\nu d} - \delta_\nu^p \delta_b^a \delta_c^d e_{\mu d}) \\ &\quad \times \delta \omega_p^{bc} \end{aligned}$$

$$\Rightarrow \delta_\mu \omega^{ab} e_{\nu b} - \delta \omega^{ab} e_{\mu b}$$

$$\therefore \frac{1}{2} \bar{E} \gamma^\alpha R_{\mu\nu}(\alpha) + 2 \delta_\mu \omega_{\nu}^{ab} e_{\alpha b} = 0$$

We must lower the α index w/ $\eta_{\alpha\beta}$

$$\frac{1}{2} \bar{E} \gamma_\alpha R_{\mu\nu}(\alpha) + 2 \delta_\mu \omega_{\nu}^{ab} \hat{e}_{\alpha b} = 0$$

$$\cancel{\frac{1}{2} \bar{E} \gamma_\mu R_{\nu\rho}(\alpha) + 2 \delta_\mu \omega_{[\nu} \hat{\mu}^{ab} e_{\rho]b} = 0}$$

Notation $\delta_\mu \omega \rightarrow \omega$

$$\frac{1}{2} \bar{E} \gamma_\mu R_{\nu\rho} + 2 \omega_{[\nu} \hat{\mu}^{ab} e_{\rho]b} \quad (a)$$

$$-\frac{1}{2} \bar{E} \gamma_\nu R_{\rho\mu} - 2 \omega_{[\mu} \hat{\rho}^{ab} e_{\nu]b} \quad (b)$$

$$+\frac{1}{2} \bar{E} \gamma_\rho R_{\mu\nu} + 2 \omega_{\rho} \hat{\nu}^{ab} e_{\mu b} \quad (c)$$

$$\omega_{\nu\mu}^{ab} e_{\rho b} - \omega_{\rho\mu}^{ab} e_{\nu b}$$

$$\text{note } \omega_{\nu\mu}^\lambda e_\lambda^b \eta_{ba} e_\rho^a = \omega_{\nu\mu}^\lambda g_{\lambda\rho}$$

$$(\omega_{\nu\mu\rho} - \omega_{\rho\mu\nu})$$

$$= \omega_{\mu\rho\nu} + \omega_{\nu\rho\mu} = +\omega_{\mu\nu\rho} - \omega_{\nu\rho\mu}$$

$$\omega_{\rho\nu\mu} - \omega_{\mu\nu\rho}$$

$$\therefore 2\omega_{\rho\nu\mu} + \frac{1}{2} \bar{E} (\gamma_\mu R_{\nu\rho} - \gamma_\nu R_{\rho\mu} + \gamma_\rho R_{\mu\nu}) = 0$$

Re-label $\rho \leftrightarrow \mu$

$$2\omega_{\mu\nu\rho} + \frac{1}{2} \bar{E} (\gamma_\mu R_{\rho\nu} - \gamma_\nu R_{\mu\rho} + \gamma_\rho R_{\nu\mu}) = 0$$

$$g^{\nu\lambda} g^{\rho\sigma} () + \frac{1}{2} \bar{E} (\gamma_\mu R^{\sigma\lambda} - \gamma^\lambda R_\mu^\sigma + \gamma^\sigma R_\mu^\lambda) = 0$$

$$2 e_a^\lambda e_b^\sigma \omega_{\mu}^{ab} + \frac{1}{2} \bar{E} (e_a^\lambda e_b^\sigma \gamma_\mu R^{ba} - e_a^\lambda e_b^\sigma \gamma^\mu R_{\mu}^{ba})$$

$$= \frac{1}{2} \bar{E} (\gamma^\mu R_\mu^\lambda - \gamma^\lambda R_\mu^\mu)$$

$$2\omega_\mu^{ab} + \frac{1}{2}\bar{\epsilon}(\gamma_\mu R^{ba} - \gamma^a R_\mu{}^b + \gamma^b R_\mu{}^a)$$

$$2\omega_\mu^{ba} + \frac{1}{2}\bar{\epsilon}(\gamma_\mu R^{ab} - \gamma^b R_\mu{}^a + \gamma^a R_\mu{}^b) = 0$$

$$\Rightarrow 2\omega_\mu^{ab} = \frac{1}{2}\bar{\epsilon}(\gamma_\mu R^{ab} + \gamma^a R_\mu{}^b - \gamma^b R_\mu{}^a)$$

* Extra minus sign missing

$$\delta_\mu \omega_\mu^{ab} = -\frac{1}{4}\bar{\epsilon}(\gamma_\mu R^{ab}) - \frac{1}{2}\bar{\epsilon}\gamma^{[a} R_\mu{}^{b]}$$

Thus, we must have: $\delta_{\text{gauge}} \omega = \frac{1}{2}\bar{\epsilon}\gamma^{ab}\phi_\mu$

why? $\rightarrow$ The formalism is such that we only write the term in $\delta\omega_\mu^{ab} \propto \bar{\epsilon}$

$$\delta_Q(\epsilon)\omega_\mu^{ab}(e,b,\psi) = \frac{1}{2}\bar{\epsilon}\gamma^{ab}\phi_\mu - \frac{1}{2}\bar{\epsilon}\gamma^{[a} R_\mu{}^{b]} - \frac{1}{4}\bar{\epsilon}\gamma_\mu R^{ab}$$

* We can use the third constraint $\gamma_\mu R_{\nu\rho} = 0$ to simplify this expression

$$\delta_Q(\epsilon)\omega_\mu^{ab}(e,b,\psi) = \frac{1}{2}\bar{\epsilon}\gamma^{ab}\phi_\mu - \frac{1}{2}\bar{\epsilon}\gamma_\mu R^{ab}(Q)$$

* We define the modified curvature:

$$\hat{R}_{\mu\nu}^A := R_{\mu\nu}^A - 2B_{[\mu}^B M_{\nu]} B^A$$

$$\hat{R}_{\mu\nu}(\psi^{ab}) = \hat{R}_{\mu\nu}^{ab} + 8\int_{[\mu}^{[a} e_{\nu]}^{b]}$$

$$\hat{R}_{\mu\nu}^{ab} = R_{\mu\nu}^{ab} + \bar{\psi}_{[\mu} \gamma^{[a} R_{\nu]}^{b]}(0) + \frac{1}{2} \bar{\psi}_{[\mu} \gamma_{\nu]} R^{ab}(0)$$

$$R_{\mu\nu}^{ab} = 2 \partial_{[\mu} \omega_{\nu]}^{ab} + 2 \omega_{[\mu}^a c \omega_{\nu]}^{cb} - \bar{\psi}_{[\mu} \gamma^{ab} \psi_{\nu]}$$

* Solving the second & third constraint:

$$e_b^\nu \hat{R}_{\mu\nu}(\mu^{ab}) = 0 \quad (2), \quad \gamma^\mu R_{\mu\nu}(a) = 0 \quad (3)$$

or $\gamma_{[\mu} R_{\nu\rho]} = 0$

$$(2): e_b^\nu \hat{R}_{\mu\nu}^{ab} + 8 e_b^\nu f_{[\mu}^{[a} e_{\nu]}^{b]} = 0$$

$$\hat{R}_\mu^a + 2 e_b^\nu (f_{[\mu}^a e_{\nu]}^b - f_{[\mu}^b e_{\nu]}^a) = 0$$

$$\Rightarrow \hat{R}_\mu^a + 2 e_b^\nu (f_\mu^a e_\nu^b - f_\nu^a e_\mu^b - f_\mu^b e_\nu^a + f_\nu^b e_\mu^a) = 0$$

$$\hat{R}_\mu^a + 2 (f_\mu^a D - f_\mu^a - f_\mu^a + f_\mu^b e_\nu^a) = 0$$

Solving for f_μ^b :

$$\hat{R}_\mu^a + 2(fD - f - f + fD) = 0$$

$$\hat{R}_\mu^a + 4f(D-1)$$

$$f = -\frac{\hat{R}_\mu^a}{4(D-1)}$$

$$\hat{R}_\mu^a + 2f_\mu^a(D-2) - \frac{\hat{R}_\mu^a}{(D-1)} = 0$$

$$f_\mu^a = -\frac{\hat{R}_\mu^a}{2(D-2)} + \frac{\hat{R}_\mu^a}{2(D-2)(D-1)} \quad \leftarrow \text{x2?}$$

will also have $+\frac{1}{4} \hat{R}_\mu^a(T)$

* Not sure why the factors of $\sim D$

are set to $D=4$ above.

$$\phi_m = -\frac{1}{2} \gamma^\nu R'_{\nu m}(Q) + \frac{i}{2} \gamma_m \gamma^{ab} R'_{ab}(Q)$$

where

$$R_{mn}(Q) = R'_{mn}(Q) - 2 \gamma_{[m} \phi_{n]}$$

$$R'_{mn}(Q) = 2 \left(\partial_{[m} \phi_{n]} + \frac{1}{2} \epsilon_{[m} \phi_{n]} - \frac{3}{2} i A_{[m} \gamma_{n]} + \frac{1}{4} \omega_{[m}^{ab} \gamma_{n]} \epsilon_{ab} \right) \phi_{[n]}$$

??

$$\text{Note: } f_m^\mu = -\frac{1}{2} \hat{R} = -\frac{1}{2} (R(\omega) - \bar{\psi}_a \gamma^{ab} \phi_b)$$

$$\gamma^\mu \phi_m = -\frac{1}{2} \gamma^{\mu\nu} R'_{\nu m}(Q)$$

$$\gamma^{ab} \phi_b = -\frac{1}{4} \gamma^{abc} R_{bc}(Q)$$

Super Conf. trans. rules of a chiral mulf.

* Global susy: Q transforms as a Maj. weyl.

→ Previous Q transformations are valid w

(i) $\bar{E}Q \rightarrow E(X)$ i.e. a f^u of the spacetime.

(ii) Derivatives of the field content of chiral mulf

$$\partial_\mu Z \text{ \& } \partial_\mu P_\mu X \rightarrow D_\mu Z, D_\mu P_\mu X$$

Where D_μ is the super covariant derivative

$$D_\mu := \partial_\mu - B_\mu^A T_A \leftarrow \text{Generators of Conf(MD)}$$

• First try to understand how Dilatations act in this setting.

⇒ Finding the can. weight h for each field of chiral mulf.

ex: Z : has c.v. h .

$$[D, Q] = \frac{1}{2} Q$$

$$\Leftrightarrow [D, Q]Z = DQZ - QDZ = D(QZ) - h(QZ)$$

$$\delta Z = \frac{1}{\sqrt{2}} \bar{E} P_L \chi$$

$$\Rightarrow \delta Z = [EQ, Z] \Rightarrow QZ = \frac{1}{\sqrt{2}} P_L \chi$$

$$\frac{1}{\sqrt{2}} [D P_L \chi - h P_L \chi] = \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} P_L \chi \right)$$

$$\Leftrightarrow D P_L \chi = (h + \frac{1}{2}) P_L \chi$$

$\therefore P_L \chi$ (the susy partner of Z) has conf. weight $\tilde{h} = h + \frac{1}{2}$

- Q applied to an element of the chiral mult. raises its conf. weight by $\frac{1}{2}$.

$\rightarrow$ Further, note, the weight of $\bar{E} P_L \chi$ must be that of Z , i.e. h . or:

$$DZ = h \Leftrightarrow D(\bar{E} P_L \chi) = h \bar{E} P_L \chi + h_\chi \bar{E} P_L \chi$$

$$\Rightarrow h = h_{\bar{E}} + h_\chi$$

However, the previous exercise has shown that $h_\chi = h + \frac{1}{2}$

$$\therefore h = h_{\bar{E}} + (h + \frac{1}{2}) \Leftrightarrow \boxed{h_{\bar{E}} = -\frac{1}{2}}$$

* The conf. weight of the susy transformation parameters ϵ have conf. weight $= -\frac{1}{2}$. Q has $\frac{1}{2}$

$\rightarrow$ Naturally, we would find, for superconf.

$$[D, S] = -\frac{1}{2} S$$

$$S_{s.c.} = \mathcal{M} S \Leftrightarrow \boxed{h_{\mathcal{M}} = \frac{1}{2}}$$

For a chiral mult. ($D_i \Phi = 0$) the auxiliary term thus has weight:

$$\delta F = \frac{1}{\sqrt{2}} \bar{E} \partial P_L \chi$$

Recall that $\partial_\mu Z$ has conf. weight $h + 1$. So the terms in $\bar{E} P_L \chi$ leave h while $\chi^m \partial_\mu$ is $+1$

$$DF = h_F F \Leftrightarrow h_F = h + 1$$

$$\left(\begin{array}{l} Z: h \\ P_L \chi: h + 1/2 \end{array}, F: h + 1 \right)$$

• T_R transformations: Define a chiral weight c s.t.

$$\delta_{T_R} Z := i c \lambda_T Z$$

and similarly for all other fields.

$$[T, Q] = -\frac{3}{2} i \gamma_* Q \quad \text{and} \quad [T, S] = \frac{3}{2} i \gamma_* S$$

These imply that $P_L \psi$ (or $\bar{\psi} P_L$) has chiral weight $3/2$

$$[T, Q] Z = T P_L \chi - Q(T Z) = -\frac{3}{2} i \gamma_* (P_L \chi)$$

$$\Rightarrow T(P_L \chi) - Q(i c Z) = -\frac{3}{2} i \gamma_* P_L \chi$$

$$T(P_L \chi) = i c P_L \chi - \frac{3}{2} i \gamma_* P_L \chi$$

$$= i(c - \frac{3}{2} \gamma_*) P_L \chi$$

$$\left(\begin{array}{l} \gamma_* P_L = \frac{1}{2} \gamma_* (1 - \gamma_*) = -P_R \\ \Rightarrow = i c P_L \chi + i \frac{3}{2} P_R \chi \end{array} \right.$$

$\Leftrightarrow$ (?? How) $\Rightarrow P_L \psi$ and $\bar{\psi} P_L$ have chiral weight $3/2$ while $P_R \psi$ has chiral weight $-3/2$

$$\text{Recall, } \delta_T = \lambda_T T$$

$$\Rightarrow [T, Q] = -\frac{3}{2} i \gamma_* Q$$

$$\text{or } \lambda_T \bar{\psi} [T, Q] Z = -\frac{3}{2} i \lambda_T (\bar{\psi} \gamma_*) Q Z$$

$$[\delta_T, \delta_Q] Z = \delta_T \delta_Q Z - \delta_Q \delta_T Z$$

$$= \delta_T (\frac{1}{\sqrt{2}} \bar{\psi} P_L \chi) - \delta_Q (i c \lambda_T Z)$$

$$= \frac{1}{\sqrt{2}} \delta_T (\bar{\psi} P_L \chi) - i c \lambda_T \delta_Q Z$$

$$\frac{1}{\sqrt{2}} \delta_T (\bar{\psi} P_L \chi) - i c \lambda_T \delta_Q Z = \frac{3}{2} i \lambda_T (\gamma_*) \frac{1}{\sqrt{2}} \bar{\psi} P_L \chi$$

$$\delta_T (\bar{\psi} P_L \chi) - i c \lambda_T (\bar{\psi} P_L \chi) = \frac{3}{2} i \lambda_T \gamma_* \bar{\psi} P_L \chi$$

$$\delta_T (\bar{\psi} P_L \chi) = i \lambda_T (c \bar{\psi} P_L \chi + \frac{3}{2} \gamma_* \bar{\psi} P_L \chi)$$

$$\hookrightarrow \delta_T(\bar{E}) P_L \chi + \bar{E} \delta_T(P_L \chi) = \delta_T(\bar{E}) P_L \chi + i \lambda_T c_\chi \bar{E} P_L \chi$$

however $c_\chi = c + 1/2$

$$\therefore i c \lambda_T \bar{E} P_L \chi + i \lambda_T (c + 1/2) \bar{E} P_L \chi = i c \lambda_T \bar{E} P_L \chi + 1/2 i \lambda_T \bar{E} P_L \chi$$

$$(c + 1/2) \bar{E} P_L \chi = 1/2 \bar{E} P_L \chi$$

($\rightarrow$ Not too sure about this $1/2$)

$$\left\{ \begin{array}{ll} P_L \epsilon \text{ \& } \bar{\epsilon} P_L : & c = 3/2 \\ P_R \epsilon \text{ \& } P_R \bar{\epsilon} : & c = -3/2 \\ P_L \eta : & c = -3/2 \\ P_R \eta : & c = 3/2 \end{array} \right\}$$

These are Chiral
Weights and NOT
Conformal weights.

Claim: The chiral weights of the fields and the parameters require that Z is invariant under S supersymmetry.

Proof: Since Q in $\delta_s = \bar{\epsilon} S$ has weight $+1/2$, Z

can only transform into a field of weight $h - 1/2$:

$$[D, S] Z = -1/2 S Z$$

$$h_S S Z - h S Z = -1/2 S Z \Rightarrow \boxed{h_S = (h - 1/2)}$$

- Recall, the Covariant derivative D_a carries a weight 1

$\therefore$ Only possibility is $\boxed{\delta_s Z = 0}$

* We can use this fact to relate the chiral weight & the Conf. weight

Start by looking at $[Q, S] Z$

Recall that, for a supersymmetry transf.:

$$\delta(\epsilon_1) \delta(\epsilon_2) = ((\bar{\epsilon}_1)^\alpha Q_\alpha) ((\bar{\epsilon}_2)^\beta Q_\beta)$$

however $(\bar{\epsilon}_1)^\alpha Q_\alpha$ is Bosonic so

$$\delta(\epsilon_1) \delta(\epsilon_2) = (\bar{\epsilon}_2)^\beta (\bar{\epsilon}_1)^\alpha Q_\alpha Q_\beta$$

$$\delta(\epsilon_2) \delta(\epsilon_1) = (\bar{\epsilon}_1)^\beta (\bar{\epsilon}_2)^\alpha Q_\alpha Q_\beta$$

$$[\delta(\epsilon_1), \delta(\epsilon_2)] = (\bar{\epsilon}_2)^\beta (\bar{\epsilon}_1)^\alpha (Q_\alpha Q_\beta) - (\bar{\epsilon}_1)^\beta (\bar{\epsilon}_2)^\alpha Q_\beta Q_\alpha \\ = (\bar{\epsilon}_2)^\beta (\bar{\epsilon}_1)^\alpha (Q_\alpha Q_\beta + Q_\beta Q_\alpha)$$

$$[\delta(\epsilon_1), \delta(\epsilon_2)] = \bar{\epsilon}_2^\beta \bar{\epsilon}_1^\alpha \{Q_\alpha, Q_\beta\}$$

$$\{Q_\alpha, Q_\beta\} = -\frac{1}{2} (\gamma^a)_\alpha^\beta P_a, \quad \{S_\alpha, S_\beta\} = -\frac{1}{2} (\gamma^a)_\alpha^\beta K_a$$

$$\{Q_\alpha, S_\beta\} = -\frac{1}{2} \delta_\alpha^\beta D - \frac{1}{4} (\gamma^{ab})_\alpha^\beta M_{ab} + \frac{1}{2} i (\gamma_5)_\alpha^\beta T_R$$

Require gct's: since $[\delta_\alpha, \delta_\beta] \sim \delta_{PM}$ then

we know that translations must be changed to gct's:

* We note, a gct has parameter ξ^μ s.t., for a real scalar field ϕ^i

$$[\delta_{gct}(\xi)] \phi^i = \xi^\mu \partial_\mu \phi^i + (\xi^\mu A_\mu^A) (t_A)^i_j \phi^j$$

* Here we have

$$\xi^\mu = \frac{1}{2} \bar{\epsilon}_2 \gamma^\mu \epsilon_1, \quad \lambda_D = \frac{1}{2} \bar{\epsilon} \epsilon, \quad \lambda_T = -\frac{1}{2} \bar{\epsilon} \gamma_5 \epsilon$$

$$\lambda_K^a = \frac{1}{2} \bar{\epsilon}_2 \gamma^a \epsilon_1, \quad \lambda^{ab} = \frac{1}{4} \bar{\epsilon} \gamma^{ab} \epsilon$$

(i) Pure Susy term

$$(\epsilon_2)_\beta (\epsilon_1)^\alpha \{Q_\alpha, Q^\beta\} = (\epsilon_2)_\beta (\epsilon_1)^\alpha (-\frac{1}{2} (\gamma^a)_\alpha^\beta P_a)$$

$$[\delta_Q(\epsilon_1), \delta_Q(\epsilon_2)] = (-\frac{1}{2} \bar{\epsilon}_1 \gamma^a \epsilon_2) P_a = (\frac{1}{2} \bar{\epsilon}_2 \gamma^a \epsilon_1) P_a \\ = \delta_{gct}(\xi^a) P_a$$

(ii) Super conf. term

$$(\eta_2)_\beta \bar{\eta}_1^\alpha \{S_\alpha, S^\beta\} = -\frac{1}{2} (\gamma^a)_\alpha^\beta K_a \Leftrightarrow [\delta_S(\eta_1), \delta_S(\eta_2)] = \delta_K(\lambda_K^a) K_a$$

$$\text{with } \lambda_K^a = \frac{1}{2} \bar{\eta}_2 \gamma^a \eta_1$$

(iii) Mixed super term:

$$[\delta_S(\eta_1), \delta_Q(\epsilon)] = \delta_D(\lambda_D) + \delta_M(\lambda^{ab}) + \delta_T(\lambda_T)$$

- The Second commutator tells us that

$$\delta_\kappa Z = 0$$

So the complex scalar in the chiral multiplet transforms trivially under SCT's as well as its supersymmetric counterpart $\delta_s Z = 0$.

$$\begin{aligned} \rightarrow [\delta_s(\eta), \delta_Q(\epsilon)] Z &= \delta_s \delta_Q Z - \delta_Q \delta_s Z \\ &= \delta_s \left(\frac{1}{\sqrt{2}} \bar{E} P_L \chi \right) \\ &= \frac{1}{\sqrt{2}} \bar{E} P_L \delta_s(\eta) \chi \end{aligned}$$

here $\bar{E} P_R$ does not appear in the result.

However this should be

$$\frac{1}{\sqrt{2}} \bar{E} P_L \delta_s(\eta) \chi = \delta_D Z + \cancel{\delta_M Z} + i \delta_T Z$$

Since we know $\delta_D Z = h Z$

$$\Rightarrow \frac{1}{\sqrt{2}} \bar{E} P_L \delta_s(\eta) \chi = h Z + i \delta_T Z$$

$$\frac{1}{\sqrt{2}} \bar{E} P_L \delta_s(\eta) \chi = \lambda_D h Z - \lambda_T c Z$$

• Split $\bar{E} = \bar{E} P_L + \bar{E} P_R$

$$\frac{1}{\sqrt{2}} (\bar{E} - \bar{E} P_R) \delta_s(\eta) \chi = \lambda_D h Z - \lambda_T c Z$$

$$\frac{1}{\sqrt{2}} \bar{E} P_L \delta_s(\eta) \chi = \frac{1}{2} (\bar{E} \eta) h Z + \frac{1}{2} \bar{E} \gamma_\mu \eta (c Z)$$

$$\frac{1}{\sqrt{2}} \bar{E} P_R \delta_s(\eta) \chi = \frac{1}{2} (\bar{E} P_L + \bar{E} P_R) \eta h Z + \frac{1}{2} (\bar{E} P_L + \bar{E} P_R) \gamma_\mu \eta (c Z)$$

$$\text{LHS} = \frac{1}{2} \bar{E} P_L (\eta (h Z) + \gamma_\mu \eta (c Z)) + \frac{1}{2} \bar{E} P_R (\eta h Z - \eta c Z)$$

- No $\bar{E} P_R$ terms on LHS so we conclude

$$\left(\frac{1}{2} \bar{E} P_R \eta \right) (h Z - c Z) = 0 \Leftrightarrow \boxed{c = h}$$

- Consider the left-handed Chiral terms

$$\frac{1}{\sqrt{2}} \bar{E} P_L \delta_s(\eta) \chi = \frac{1}{2} h Z \bar{E} P_L \eta + \frac{1}{2} h Z \bar{E} P_L \eta$$

$\delta P_L \chi$

$$\delta_s(n) P_L \chi = \frac{1}{2} \epsilon P_L \chi$$

$$= \frac{1}{2} P_L (\not{D} \epsilon + F \epsilon)$$

* Similarly, we can find the local Super conf transformations for the auxillary field F in $(D_a \epsilon = 0)$.

$$\begin{aligned} \bullet [\delta_s, \delta_a] \chi &= \delta_s \delta_a \chi - \delta_a \delta_s \chi \\ &= \delta_s (\not{D} \epsilon + F \epsilon) - \delta_a (\frac{1}{2} \epsilon P_L \chi) \end{aligned}$$

* NOTE: Since we are Gauging our symmetries, we must Replace $\partial_a \epsilon \mapsto D \epsilon$ (should be D_a ?)

$$\begin{aligned} \delta_s (\not{D} \epsilon + F \epsilon) - \frac{1}{2} \epsilon P_L \chi &= (\delta_s F) \epsilon - \frac{1}{2} \epsilon P_L \chi + \delta_s (\not{D} \epsilon) \epsilon \end{aligned}$$

ⓐ Lemma states $\delta_s (D_a \epsilon)$ can only depend on the values previously found.

$$\Rightarrow D_a \epsilon = \partial_a \epsilon - A_a^A T_A \epsilon$$

$$\begin{aligned} \delta (D_a \epsilon) &= \delta_s (\partial_a \epsilon) - (\delta A_a^A) T_A \epsilon \\ &= - \delta A_a^A T_A \epsilon \end{aligned}$$

$$= - \delta_s e_a^\mu P_\mu \epsilon - \delta_s \tilde{\omega}_{ab}^c M_{[ab]} \epsilon - \delta_s b_a D \epsilon - \delta_s A_a Q \epsilon$$

First term vanishes as scalars are invariant under translations

the third vanishes since $\delta_s b_a \sim \lambda_k$ so it is a SCT of

$\delta_s \epsilon = 0$. We are simply left w/ the last term which is a SUSY:

$$\gamma^\mu \delta_s (D_a \epsilon) = - \gamma^\mu (\gamma_a e_\mu^\nu) \frac{1}{2} P_L \chi = \gamma^\mu \gamma_a \chi$$

$$\begin{aligned} - (\gamma_a)_\alpha^\beta \chi_\beta &= (\gamma_a)_\alpha^\beta \chi^\gamma C_{\gamma\beta} = - (\gamma_a)_\alpha^\beta C_{\beta\gamma} \chi^\gamma \\ &= - (\gamma_a)_\alpha^\gamma \chi^\gamma = - \bar{\chi} (\gamma_a)_\alpha \end{aligned}$$

$$- \frac{1}{2} \gamma^\mu \bar{\chi} (\gamma_a)_\alpha P_L \chi$$

$$\text{ⓐ } \gamma^\mu \perp \bar{\chi} \gamma_\mu \chi$$

$$a) = \delta(\sqrt{2} \gamma^a \gamma_5 \gamma_L \chi)$$

$$\therefore [\delta_8, \delta_2] \chi$$

$$= \delta_5 F \in -\sqrt{2} h m \bar{E} P_L \chi - \frac{1}{2} \gamma^a \bar{m} \gamma_a P_L \chi \in$$

$$= \delta_D \chi + \delta_M \chi + \delta_T \chi$$

$$= (h + 1/2) (\frac{1}{2} \bar{E} m) \chi + \frac{1}{2} \bar{E} \gamma^{ab} m [\frac{1}{2} \gamma_{ab}] \chi - \frac{1}{2} \bar{E} \gamma^a m (h + 1/2) \chi$$

$$\bullet \bar{E}^\alpha (\gamma^{ab})_\alpha^\beta m_\beta (\gamma_{ab})_\gamma^\delta \chi_\delta$$

$$\rightarrow \gamma^{ab} \gamma_{ab} = \frac{1}{4} (\gamma^a \gamma^b - \gamma^b \gamma^a) (\gamma_a \gamma_b - \gamma_b \gamma_a)$$

$$= \frac{1}{4} (\gamma^a \gamma^b \gamma_a \gamma_b - \gamma^b \gamma^a \gamma_a \gamma_b - \gamma^a \gamma_b \gamma_b \gamma_a + \gamma^b \gamma_a \gamma_a \gamma_b)$$

$$= \frac{1}{2} (\gamma^a \gamma^b \gamma_a \gamma_b - \gamma^a \gamma^b \gamma_b \gamma_a)$$

$$= \frac{1}{2} (-\gamma^b \gamma^a \gamma_a \gamma_b + 2 \eta^{ab} \gamma_a \gamma_b - \gamma^b \gamma^a \gamma_a \gamma_b)$$

$$= (\eta^a{}_a - \gamma^b \gamma^a \gamma_a \gamma_b)$$

$$= (D^2 - 1) \mathbb{1}$$

$$\Rightarrow \bar{E}^\alpha (D^2 - 1) \delta_\alpha^\beta \delta_\beta^\gamma \chi_\gamma m_\beta = (\bar{E} m) (D^2 - 1) \chi$$

$$\Leftrightarrow -\frac{1}{4} (\bar{E} m) (D^2 - 1) \chi$$

$$\bullet H S = (h + \frac{1}{2} \gamma^a \gamma_a) \chi - \frac{1}{4} (\bar{E} m) (D^2 - 1) \chi + \frac{1}{2} \bar{E} \gamma^a m (h + \frac{1}{2}) \chi$$

$$\rightarrow R H S = (\delta_5 F) \in + \sqrt{2} h \bar{E} m P_L \chi - \frac{1}{2} \gamma^a \bar{m} \gamma_a P_L \chi \in$$

(b)

$$\textcircled{b} = -\frac{1}{2} (\gamma^a)_\alpha^\beta \bar{m}^\delta (\gamma_a)_\gamma^\delta (P_L \chi)_\delta \in \beta$$

$$= -\frac{D}{2} \epsilon_\alpha (\bar{m} P_L \chi)$$

$$R H S = (\delta_5 F) \in + \sqrt{2} h (\bar{E} m) P_L \chi - \frac{D}{2} \in (\bar{m} P_L \chi)$$

$$L H S = (h + \frac{1}{2}) \frac{1}{2} (\bar{E} m) \chi - \frac{1}{4} (D^2 - 1) (\bar{E} m) \chi + \frac{1}{2} \bar{E} \gamma^a m (h + \frac{1}{2}) \chi$$

$$\rightarrow M = 2^{-n} \sum_{k=0}^{\infty} \frac{1}{k!} \int_{\mu_1, \dots, \mu_k} \text{Tr}(\gamma^{\mu_1 \dots \mu_k} M)$$

$$\bar{E}^\alpha m_\beta \chi_\alpha \sim (M_\beta)^\alpha = \frac{1}{2^n} \sum_{k=0}^{\infty} \frac{1}{k!} \left(\gamma^{\mu_1 \dots \mu_k} \right)_\beta^\alpha \left(\gamma^{\mu_1 \dots \mu_k} \right)_\delta^\beta M_\delta^\alpha$$

$$= \frac{1}{2^n} (M_\delta^\delta \delta_\alpha^\alpha + (\gamma_{\mu_1})_\alpha^\beta (\gamma^{\mu_1})_\gamma^\delta M_\delta^\alpha + \frac{1}{2} (\gamma_{\mu_1 \mu_2})_\alpha^\beta (\gamma^{\mu_1 \mu_2})_\gamma^\delta M_\delta^\alpha$$

$$+ \frac{1}{3!} (\gamma_{\mu_1 \mu_2 \mu_3})_\alpha^\beta (\gamma^{\mu_1 \mu_2 \mu_3})_\gamma^\delta M_\delta^\alpha + \dots + \frac{1}{k!} (\gamma_{\mu_1 \dots \mu_k})_\alpha^\beta (\gamma^{\mu_1 \dots \mu_k})_\gamma^\delta M_\delta^\alpha$$

$$\gamma^{\mu_1 \dots \mu_k} = i^k \epsilon^{\mu_1 \dots \mu_k} \gamma_5, \quad \gamma^{\mu_1 \mu_2 \mu_3} = \epsilon^{\mu_1 \mu_2 \mu_3} \gamma_{\mu_4}$$

Start here: (Explain γ via duality)

$$\gamma^{\mu_1 \dots \mu_k} \gamma_{\mu_1 \dots \mu_k} = \frac{D!}{(D-2k)!} \mathbb{1} = D(D-1)$$

$$\Rightarrow \delta_S(M)F = \sqrt{2}(1-h)\bar{\eta}P_L\chi$$

Complete Superconformal trans. for the chiral multiplet
(w/o get's ξ Lorentz)

$$\delta Z = (h\lambda_0 + i h\lambda_T)Z + \frac{1}{\sqrt{2}}\bar{\epsilon}P_L\chi + 0$$

$$\delta P_L\chi = [(h+\frac{1}{2})\lambda_0 + (h-\frac{3}{2})i\lambda_T]P_L\chi + \frac{1}{\sqrt{2}}P_L(\not{D}Z + F)\epsilon + \sqrt{2}hZP_L\eta$$

$$\delta F = [(h+1)\lambda_0 + (h-3)i\lambda_T]F + \frac{1}{\sqrt{2}}\not{D}P_L\chi + \sqrt{2}(1-h)\bar{\eta}P_L\chi$$

- Dilations (δ_D)
- Super symmetry (δ_a)
- R-sym (δ_{T_R})
- Super Conformal (δ_S)

* For chiral superfields we find that the R-symmetry $U(1)_R$ charge is completely determined by the Conf. weight of the field in the multiplet.
($\sim$ up to normalization of the U_1 matrix)

* Any supercovariant action will also be T_R -symmetric (even locally)
 $\rightarrow$ Superconf. cov. der.:

$$D_\mu Z = e_a^\mu (\partial_\mu Z - h b_\mu - i A_\mu - \frac{1}{\sqrt{2}} \bar{\Psi}_\mu P_L \chi)$$

Note: there is no Lorentz term as Z is a scalar
 $P_L\chi$, however, contains Lorentz/Poincare terms, i.e. $\sim \omega_m^{ab}$
spin connection terms are also present:

$$D_\mu P_L\chi = e_a^\mu P_L \left[\partial_\mu + \frac{1}{4} \omega_m^{bc} \gamma_{bc} - (h+\frac{1}{2})b_\mu - i(h-\frac{3}{2})A_\mu \right] \chi - \frac{1}{\sqrt{2}}(\not{D}Z + F)\psi_\mu - \sqrt{2}hZ\phi_\mu$$

• In global susy F transforms as a total der. Therefore FF^* is the appropriate 'Lagrange multiplier.'

$\hookrightarrow$ In a theory of gravity we should expect $\sim e F$

Since the measure has changed from $M^D \rightarrow M^D$

- e : Conf. weight = $-4 = w_e$

So F should carry weight $+4$. From above we

see it carries $\boxed{w+1} \Leftrightarrow \boxed{w=3}$

- For this choice we also find that F becomes invariant under R -symmetry as well

$\int d^4x e F$: NOT invariant under SCT's as $\delta_\alpha F$ & $\delta_\beta F$ lower/rise the weight/field.

The modified integral becomes

(Master Formula)

$$S_F = \int d^4x e \operatorname{Re} \left\{ F + \frac{1}{\sqrt{2}} \bar{\Psi}_m \gamma^m P_L \chi + \frac{1}{2} z \bar{\Psi}_m \gamma^{mn} P_R \psi_n \right\}$$

(Both $\operatorname{Re}\{\cdot\}$ & $\chi_m\{\cdot\}$ give SC. invariant Actions for a chiral multiplet of Conf. weight $w=3$.)

Proof: We prove that S_F is invariant under δ_S :

$$\delta_S S_F = \int d^4x \delta_S (e \operatorname{Re}\{\cdot\})$$

$$= \int d^4x (\delta_S e \operatorname{Re}\{\cdot\} + e \operatorname{Re}\{\underbrace{\delta_S F}_a + \underbrace{\frac{1}{\sqrt{2}} \delta_S (\bar{\Psi}_m \gamma^m P_L \chi)}_b + \underbrace{\frac{1}{2} \delta_S (z \bar{\Psi}_m \gamma^{mn} P_R \psi_n)}_d\})$$

$$\text{w/ } \delta_S e = 0$$

$$\textcircled{a}: \delta_S F = \sqrt{2}(1-w) \bar{\pi} P_L \chi \Rightarrow \operatorname{Re}\{\delta_S F\} = \sqrt{2}(1-w) \bar{\pi} P_L \chi \quad \leftarrow$$

$$\textcircled{b} \quad \frac{1}{\sqrt{2}} \delta_S \bar{\Psi}_m \gamma^m P_L \chi + \frac{1}{\sqrt{2}} \bar{\Psi}_m \gamma^m P_L (\sqrt{2} w z \pi) \\ \hookrightarrow = -\bar{\pi} e_a^\mu \gamma^a (\frac{1}{\sqrt{2}} P_L \chi)$$

$$\textcircled{d} \quad \frac{1}{2} z (-\bar{\pi} e_a^\mu \gamma_a) \gamma^{mn} P_R \psi_n - \frac{1}{2} z \bar{\Psi}_m \gamma^{mn} P_R (\gamma_n \pi)$$

$$\hookrightarrow \int d^4x e \operatorname{Re} \left\{ \sqrt{2}(1-w) \bar{\pi} P_L \chi - \frac{1}{2} \bar{\pi} \gamma_m \gamma^m P_L \chi + w z \bar{\Psi}_m \gamma^m P_L \pi - \frac{1}{2} z \bar{\pi} \gamma_m \gamma^{mn} P_R \psi_n - \frac{1}{2} z \bar{\Psi}_m \gamma^{mn} P_R (\gamma_n \pi) \right\} \\ = \int d^4x e \operatorname{Re} \left\{ \sqrt{2}(1-w) \bar{\pi} P_L \chi - 0 \bar{\pi} P_L \chi + w z \bar{\Psi}_m \gamma^m P_L \pi - \frac{1}{2} z \bar{\pi} (0-1) \gamma^\nu P_R \psi_\nu - \frac{1}{2} z \bar{\Psi}_m \gamma^{mn} P_R \gamma_n \pi \right\}$$

$$\left\{ \begin{aligned} \{\gamma_\mu, \gamma_\nu\} = 0 &\Rightarrow \gamma^\mu P_\pm = \frac{1}{2} \gamma_\mu (1 \pm \gamma_5) = \frac{1}{2} (\gamma_\mu \mp \gamma_\mu \gamma_5) \\ \gamma^\mu P_\pm &= \frac{1}{2} (1 \mp \gamma_5) \gamma_\mu = P_\mp \gamma_\mu \end{aligned} \right.$$

$$\int d^4x \operatorname{Re} \left\{ \sqrt{2} (1-h) \bar{\chi} P_L \chi - D \bar{\chi} P_L \chi + h \bar{Z} \bar{\Psi}_m \gamma^\mu P_L \chi \right. \\ \left. - \frac{1}{2} (D-1) \bar{\chi} \gamma^\mu P_R \Psi_m - \frac{1}{2} \bar{Z} \bar{\Psi}_m \gamma^\mu (D-1) P_L \chi \right\}$$

Terms cancel

$$\int d^4x \operatorname{Re} \left\{ \sqrt{2} (1-h) \bar{\chi} P_L \chi - D \bar{\chi} P_L \chi + h \bar{Z} \bar{\Psi}_m \gamma^\mu P_L \chi \right\}$$

$\rightarrow$ Since we are building a $D=4$ dim theory w/ Z of Conf. weight $h=3$ we find

$$\int d^4x \operatorname{Re} \left\{ \sqrt{2} (-2) \bar{\chi} P_L \chi + \frac{4}{\sqrt{2}} \bar{\chi} P_L \chi + 3 \bar{Z} \bar{\Psi}_m \gamma^\mu P_L \chi \right\}$$

$\rightarrow$ third term cancels (cancelled w/ $\frac{4}{\sqrt{2}}$ term) $\square$

* A chiral mult. is determined by its lowest component both in global SUSY and in the superconf setting.

$\Rightarrow$ If one field transforms as Z w/ χ replaced by some arbitrary spinor, then the superconf. algebra determines that this χ should transform as the χ above for some F .

Superconf. action of the chiral Multiplet:

Idea: Generalize

$$S = \int d^Dx \left(\frac{1}{2} g_{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \frac{D-2}{8(D-1)} R \phi^2 \right)$$

which is a purely gravitational theory coupled to scalars

• $\phi(x)$ is not physical

$\rightarrow$ Need a scalar field w/ weight $h=1$ (as in S above w/ ϕ)

$\rightarrow$ We can obtain " ϕ " through Z in the multiplet of

$$\{Z, P_L \chi, F\} \text{ with } \boxed{h=1}.$$

Steps:

(i) If Φ is a chiral multiplet ($D_{\alpha}\Phi=0$) then the anti-chiral multiplet $\Phi^{\dagger}$ s.t. $D_{\alpha}\Phi^{\dagger}=0$ has the field content $\{\bar{Z}, \bar{P}_R, \bar{F}\}$

$\rightarrow$ The fields $\{Z' := \bar{F}, P_L \chi' := \partial P_R \chi, F' := D\bar{Z}\}$ are also components of a chiral mult.

The weyl weight of $\bar{F}$ is $h_{\bar{F}} = 2$ since $h_{\chi} = 3/2$

(ii) Consider the product multiplet:

$$\{Z\bar{F}, P_L(Z\partial\chi + \bar{F}\chi), F\bar{F} + ZD\bar{Z} - \bar{\chi}P_L\partial\chi\}$$

This is from $\Phi^i \Phi^j \rightarrow D_{\alpha}(\Phi^{(1)}\Phi^{(2)}) = 0$

where Z^i of Φ^2 is

The lowest weight state is $h=3$ while the highest weight state has $h=4$

(iii) The Action function:

$$S_F = \int d^4x \operatorname{Re} \left\{ F + \frac{1}{12} \bar{\psi}_m \gamma^m P_L \chi + \frac{1}{2} Z \bar{\psi}_m \gamma^{mn} P_R \psi_n \right\} e$$

gives a locally superant. mv. action for a chiral Mult.

whose weyl weight is $h_F = 4$

Q: why? recall that e carries weyl weight $h_e = -4$

Since $e = \frac{1}{4!} \epsilon^{a_1 \dots a_4} e_{a_1}^{m_1} e_{a_2}^{m_2} \dots e_{a_4}^{m_4} \epsilon_{m_1 \dots m_4}$ it carries

D values of e_a^m so (eF) must have weight 0.

$$S_F = \int d^4x \int d^4\theta \Phi^i \Phi^i$$

$$\text{s.t. } \Phi(x^m, \bar{\theta}) = Z(y^n) + \bar{\theta} P_L \chi + \bar{\theta} \bar{\theta} F(y)$$

$$\text{and } D_{\alpha}\Phi = 0$$

$\rightarrow \int d^4\theta \Phi^i \Phi^i$ must be modified in such a way that the "F-term" yields S_F above.

* Assume $h_F=4$ is the Chiral superfield to be the local superant version of the product multiplet.

→ Obtain the superconf version of the chiral mult whose lowest comp. is $\bar{F}$.

$$\delta F = [(h+1)\lambda_0 + i(h-3)\lambda_1]F + \frac{1}{\sqrt{2}}\bar{\epsilon}\not{D}P_L\chi + \sqrt{2}(1-h)\not{\epsilon}P_L\chi$$

Simplifies if $h=1$, removing the last term, ridding us of $\not{\epsilon}$.

→ Then δF is similar to that of $\delta \bar{Z}$ at $h_{\bar{Z}}=2$

$$\delta \bar{Z} = [2\lambda_0 - i2\lambda_1]\bar{Z} + \frac{1}{\sqrt{2}}\bar{\epsilon}\not{P}_L\chi'$$

$$\rightarrow \delta F|_{h=1} = [2\lambda_0 - 2i\lambda_1]F + \frac{1}{\sqrt{2}}\bar{\epsilon}\not{D}P_L\chi$$

• From the complex conjugate we identify

$$\boxed{P_L\chi' = \not{D}P_R\chi}$$

We can now use ^{the} Q-supersymmetry transform of $P_L\chi'$ to find the "top" F' component of the multiplet.

$$\boxed{Z' = \bar{F}, \quad P_L\chi' = \not{D}P_R\chi, \quad F' = \square^C \bar{Z}}$$

In super space this is simply $\boxed{D_{\dot{\alpha}} D_{\dot{\beta}} \Phi^{\dagger}}$

$$S_F = \int d^4x \int d^4\theta D^{\dot{\alpha}} D_{\dot{\alpha}} \Phi^{\dagger}$$

→ Super conformal multiplet w/ $h=2$

Recap: $\{(Z, P_L\chi, F), h_{\bar{Z}}=1\}$ i.e. Z is a scalar field

→ $\{(\bar{F}, \not{D}P_R\chi, \square^C \bar{Z}), h=2\}$

→ The product of these two $\{(Z\bar{F}, \dots), h=3\}$

→ The superconf. mult. for $\Phi^{\dagger}$ must be realised differently.

→ components must be similar to $\delta F, \delta P_L\chi, \dots$

Recall

$$\square^C \phi := \not{m}^b \not{D}_a D_b \phi, \quad D_a \phi := e_a^m (\partial_m - h b_m) \phi$$

to compute $D_a(D_b \chi)$ take $\delta(D_b \chi)$ and replace coeff. by their respective generators

$$D_a \chi = e_a^\mu (\partial_\mu \chi - h b_\mu \chi - i h A_\mu^{(R)} \chi - \frac{1}{\sqrt{2}} \bar{\Psi}_\mu P_L \chi)$$

$\therefore$ To find $D^a \chi$ we must take $\delta(D_a \chi)$... Fuck ...

$$\begin{aligned} \delta(D_a \chi) &= \delta e_a^\mu (1 + e_a^\mu (\partial_\mu \chi - h b_\mu \chi - i h A_\mu^{(R)} \chi - \frac{1}{\sqrt{2}} \bar{\Psi}_\mu P_L \chi)) \\ &= (\lambda_a^b e_b^\mu + \lambda_0 e_a^\mu) (1 + e_a^\mu [\lambda_0 + i h \lambda_T] \partial_\mu \chi + \frac{1}{\sqrt{2}} \bar{\Psi}_\mu P_L \chi \\ &\quad - h \lambda_\mu^b e_{\mu b} \chi - h b_\mu (h \lambda_0 + i h \lambda_T) \chi - h \frac{1}{\sqrt{2}} \bar{\Psi}_\mu P_L \chi \\ &\quad (-i h (h \lambda_0 + i h \lambda_T) \chi - i h \frac{1}{\sqrt{2}} \bar{\Psi}_\mu P_L \chi) A_\mu^{(R)} \\ &\quad + \frac{1}{\sqrt{2}} \bar{\Psi}_\mu \gamma_0 P_L \chi - \frac{1}{\sqrt{2}} \bar{\Psi}_\mu [(h + \frac{1}{2}) \lambda_0 + (h - \frac{3}{2}) i \lambda_T] P_L \chi + \frac{1}{\sqrt{2}} P_L (D \chi + F) \epsilon \\ &\quad + \sqrt{2} h \chi P_L \chi] \end{aligned}$$

$$= \lambda_a^b D_b \chi + \lambda_0 D_a \chi + e_a^\mu h \lambda_0 \partial_\mu \chi - h b_\mu \lambda_0 \chi - i \lambda_0 \chi A_\mu^{(R)} - \frac{3}{2\sqrt{2}} \bar{\Psi}_\mu \lambda_0 P_L \chi$$

$$\Rightarrow \lambda_a^b D_b \chi + 2 \lambda_0 D_a \chi + \dots$$

$$\boxed{D^a \chi = \eta^{ab} D_b \chi = e_a^\mu (\partial_\mu D_a \chi - 2 b_\mu D_a \chi + h_{\mu a b} D^b \chi + 2 f_{\mu a} \chi - i A_\mu^{(R)} D_a \chi - \frac{1}{\sqrt{2}} \bar{\Psi}_\mu P_L D_a \chi + \frac{1}{\sqrt{2}} \bar{\Psi}_\mu \gamma_a P_L \chi)}$$

✓ To check the Q-susy of $P_L \chi$, one needs the transformation of $\gamma^a D_a P_R \chi$. Since $P_L \chi' = \not{D} P_R \chi$

$$\Rightarrow \delta_Q P_L \chi' = \delta_Q (\not{D} D_a P_R \chi)$$

Note since there are no terms "underlined" in the expansions so $\delta(P_a) = 0$ and we simply have

$$\delta_Q (P_L \chi') = \gamma^a D_a (P_R (\frac{1}{\sqrt{2}} (D \chi + F) \epsilon)) = \frac{1}{\sqrt{2}} \not{D} P_R (D \chi + F) \epsilon$$

$$\not{D} \not{D} = \square^c + \gamma^{ab} D_a D_b$$

$\rightarrow$ Ignore second term, as γ^{ab} is not in the final result of a susy transformation

$$\delta Q \chi = (D^\mu Z + D^\mu \chi) \epsilon$$

$$\Rightarrow F' = D^\mu Z$$

The product of $(Z, P_L \chi, F)$ and $(\bar{F}, \bar{\chi} P_R, D^\mu Z)$

$$\begin{cases} Z(Z\bar{F}) = Z\bar{F} \\ P_L \chi(Z\bar{F}) = P_L \chi \bar{F} + Z \not{D} P_R \chi \\ F(Z\bar{F}) = F\bar{F} + Z D^\mu Z - \bar{\chi} P_L \not{D} \chi \end{cases}$$

• Last term leads to the S_F term in global SUSY.

* Scalars appear w/ the superconf. extension of the d'Alembertian.

We substitute those into $S_F = \int d^4x e \text{Re} \{ F\bar{F} + \frac{1}{2} \bar{\Psi}_m \delta^{\mu\nu} P_L \chi + \frac{1}{2} Z \bar{\Psi}_m \gamma^{\mu\nu} P_R \psi_\nu \}$

$$S_{S_0} = - \int d^4x e \text{Re} \{ F\bar{F} + Z D^\mu Z - \bar{\chi} P_L \not{D} \chi + \frac{1}{2} \bar{\Psi}_m \delta^{\mu\nu} (P_L \chi \bar{F} + Z \not{D} P_R \chi) + \frac{1}{2} Z \bar{F} \bar{\Psi}_m \gamma^{\mu\nu} P_R \psi_\nu \}$$

* Expanding $\not{D}$ shows that only the $F\bar{F}$ term will remain in the Action.

Gauge-fixing: Must Gauge all symm. which are not part of the Super-Poincaré: (K_a, D, T_R, S)

(i) SCT's: Since $\delta_\chi(\lambda_\mu) b_\mu = +2(\lambda_\mu)_\mu$ we can use $E_{b_\mu=0}$ to go to the gauge $b_\mu = 0$

(ii) Dilatations: We fix $|Z|$ instead of $\phi = \frac{1}{K}$
 $\rightarrow$ Must also fix the T_R -Symm as well.

$$(D\text{-gauge} \& T\text{-gauge}): Z := \frac{\sqrt{3}}{K}$$

$$(iii) (S\text{-gauge}): \chi = 0$$

$$\Rightarrow D_a Z = -i A_a Z$$

$$D_a P_L \chi = e_a^\mu P_L \left[\frac{1}{2} (i K Z - F) \psi_\mu - \sqrt{2} Z \phi_\mu \right]$$

$$D^\mu Z = -i e^{a\mu} (D_a Z + \dots) = -i A^\mu Z - \frac{1}{\sqrt{6}} \bar{\psi} \gamma^\mu \psi$$

$$\therefore S_{SG} = - \int d^4x \text{Re} \{ F \bar{F} + \frac{3}{k^2} (-i \partial^n (\partial_m A_n + \omega_{mab} A^b) + 2 f_m^{\mu\nu} - A^a A_a) \} \\ + \frac{1}{2} \bar{\psi}_m \gamma^m \epsilon \{ \underbrace{e_a^\mu \rho_\mu}_{\mathcal{O}(6^3) \text{ term}} (\frac{1}{2} (i \not{A} \epsilon - F) \psi - \sqrt{2} \epsilon \phi_m) \}$$

$$\Rightarrow S_{SG} = - \int d^4x e \left[F \bar{F} + \frac{3}{k^2} (2 f_m^{\mu\nu} - A^a A_a - \frac{1}{2} \bar{\psi}_m \gamma^{\mu\nu} \psi + \frac{1}{4} i \bar{\psi}_m \gamma^{\mu\nu} \not{A}_\mu \psi) \right] \\ \text{using } f_m^{\mu\nu} = -\frac{1}{i^2} \hat{R}_{\mu\nu}^m - \frac{1}{12} (R(\omega) - \bar{\psi}_a \gamma^{ab} \psi_b) \\ \gamma^m \psi_m = -\frac{1}{6} \gamma^{\mu\nu} R'_{\mu\nu}(\omega) \\ \gamma^{ab} \psi_b = -\frac{1}{4} \gamma^{abc} R'_{abc}(\omega)$$

$\therefore$ Superconformal Gravity: in $D=4$

$$S_{SG} = \int d^4x \left[\frac{1}{2k^2} (R(\omega(e, \psi)) - \bar{\psi}_\mu \gamma^{\mu\nu\rho} (\partial_\nu + \frac{1}{4} \omega_\nu^{ab}(e, \psi) \gamma_{ab}) \psi_\rho + 6 A^a A_a) - F \bar{F} \right]$$

* When the gravitino torsion terms are split from $\omega_\mu^{ab}(e, \psi)$ we find an extra 4-gravitino terms in $N=1, D=4$

supergravity

Recall: $D=4, N=1$ Supergravity: $S = S_2 + S_{3/2}$

$$S_2 = \frac{1}{2k^2} \int d^4x e e^{a\mu} e^{b\nu} R_{\mu\nu ab}(\omega)$$

$$S_{3/2} = -\frac{1}{2k^2} \int d^4x e \bar{\psi}_\mu \gamma^{\mu\nu\rho} D_\nu \psi_\rho$$

$$\Rightarrow \text{First order formalism } \omega_\mu^{ab} = \omega_\mu^{ab}(e) + K_\mu^{ab}$$

$$K_\mu^{ab} = -\frac{1}{2} (T_{[\mu\nu]\rho} - T_{[\nu\rho]\mu} + T_{[\rho\mu]\nu})$$

$\triangleright$ Solve for $T_{ab}{}^\nu$ and plug back into action

$$S = \frac{1}{2k^2} \int d^4x (e e^{a\mu} e^{b\nu} R_{\mu\nu ab} - \bar{\psi}_\mu \gamma^{\mu\nu\rho} D_\nu(e) \psi_\rho + \mathcal{L}_{\text{torsion}}^{(4)})$$

• F and A^a are auxiliary fields.

→ Include Supersymm. transformations: $\delta \psi_m^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_m$

ψ_m : For the gravitino we must take the gauge fixing conditions into account: $b_m = 0$, $z = \frac{\sqrt{3}}{\kappa}$, $\chi = 0$

Note: $\chi = 0$ (S-gauge) is not supersymmetric and is only super-invariant for combinations of Q & S-susy.

$$\delta P_L \chi = \left[(h + \frac{1}{2}) \lambda_0 + i(h - \frac{3}{2}) \lambda_T \right] P_L \chi + \frac{1}{\sqrt{2}} P_L (\not{\partial} z + F) \epsilon + \sqrt{2} h z P_L \eta$$

→ under Gauge: $\delta P_L \chi = \frac{1}{\sqrt{2}} P_L F \epsilon + \frac{\sqrt{6}}{\kappa} h P_L \eta$ why?

$$+ \frac{1}{\sqrt{2}} P_L \gamma^m (\partial_m z - i h A_m z)$$

$$\delta P_L \chi = \frac{1}{\sqrt{2}} P_L F \epsilon - \frac{1}{\sqrt{2}} h P_L \chi \left(\frac{\sqrt{3}}{\kappa} \right) + \frac{\sqrt{6}}{\kappa} h P_L \eta \stackrel{!}{=} 0$$

$h = 1$: $P_L \eta = i \frac{1}{2} P_L \chi - \frac{\kappa}{2\sqrt{3}} F P_L \epsilon$

$\therefore P_L \eta = \frac{1}{2} i P_L \chi \epsilon - \frac{\kappa}{2\sqrt{3}} F P_L \epsilon$ ("Decomposition law")

$$\therefore \delta \psi_m \sim \gamma_a \epsilon_m^a \left(\frac{1}{2} i \chi \epsilon - \frac{\kappa}{2\sqrt{3}} F P_L \epsilon \right)$$

$$\delta \psi_m = -\frac{1}{4} \lambda^{ab} \partial_{ab} \psi_m + \left(\partial_m + \frac{1}{4} \omega_m^{ab} \gamma_{ab} - \frac{3}{2} i A_m \gamma_T \right) \epsilon - \frac{1}{2} \gamma_a \epsilon_m^a i \chi \epsilon + \frac{\kappa}{2\sqrt{3}} \gamma_a \epsilon_m^a \bar{F} \epsilon - \frac{1}{2} \lambda_0 \psi_m + \frac{3}{2} i \lambda_T \gamma_T \psi_m$$

$\Rightarrow \delta P_L \psi_m = P_L \left(\partial_m + \frac{1}{4} \omega_m^{ab} \gamma_{ab} - \frac{3}{2} i A_m + \frac{1}{2} i \chi + \frac{\kappa}{2\sqrt{3}} \gamma_m \bar{F} \right) \epsilon$
($\lambda_T \equiv 0$)

* Must find susy transformations for A_m and F :

$$\delta A_m = \partial_m \chi_T - i \bar{\epsilon} \gamma_T \psi_m + \frac{1}{2} i \gamma_T \not{\partial} \psi_m$$

$$= -i \bar{\epsilon} \gamma_T \psi_m + \frac{1}{2} i \gamma_T \not{\partial} \psi_m$$

• We can use the lemma of cov. derivatives:

δA_m or ψ_m : Gauge field for superpoincaré supergravity
so do not consider this term:

→ look at $\bar{\epsilon}$ term: Recall

$$\psi_m = -\frac{1}{2} \gamma^{\nu} R_{\mu\nu}(\partial) + \frac{2}{12} \gamma_m \gamma^{ab} R_{ab}(\partial)$$

$$\Rightarrow \delta A_m = -i \bar{\epsilon} \gamma_T \left(-\frac{1}{2} \gamma^{\nu} \hat{R}_{\mu\nu}(\partial) + \frac{2}{12} \gamma_m \gamma^{ab} \hat{R}_{ab}(\partial) \right)$$

Where $\hat{R}_{mn}^A := r_{mn}^A - 2 B_{[m}^B M_{n]}^A$
 $\delta(\epsilon) B_m^A = \partial_m \epsilon^A + \epsilon^C B_m^B f_{BC}^A + \epsilon^B M_{mB}^A$
 $r_{mn}^A = 2 \partial_{[m} B_{n]}^A + B_n^C B_m^B f_{BC}^A$

$R_{mn}^A(Q) = 2(\partial_{[m} - \frac{3}{2} i A_{[m} \gamma_{n]} + \frac{1}{4} \omega_{[m}^{ab} \gamma_{n]} \gamma_{ab}) \psi_{n]}^A$
 $P_L \hat{R}_{mn}^A(Q) = 2 P_L (\partial_{[m} + \frac{1}{4} \omega_{[m}^{ab} \gamma_{n]} \gamma_{ab} - \frac{3}{2} i A_{[m} + \frac{1}{2} i \partial_{[m} A + \frac{4}{2\sqrt{3}} \partial_{[m} F) \psi_{n]}^A$

For F : $\delta F = \left[2\lambda_0 F + \frac{1}{12} \epsilon \gamma^\mu \left(-\frac{1}{12} (\not{D} Z + F) \psi_\mu - \sqrt{2} Z \phi_\mu \right) \right]$

ψ_μ can be omitted and $\lambda_0 = 0$

Note $\gamma^\mu \phi_\mu = -\frac{1}{6} \gamma^{\mu\nu} \hat{R}_{\mu\nu}(Q)$

$\therefore \delta F = \frac{5}{6\kappa} \epsilon P_R \gamma^{\mu\nu} \hat{R}_{\mu\nu}(Q)$

$\rightarrow$ With these susy transformations, the supergravity multiplet closes off-shell

* For any chiral theory, we can also add superpotential terms $\sim \int d^4\theta W(\Phi) \delta^2(\theta) + \int d^4\theta W^\dagger(\bar{\Phi}) \delta^2(\bar{\theta})$
 However, we must check that $\mathcal{F}$ is super conformal.

$S_F = \int d^4x \left[F W'(z) - \frac{1}{2} F P_L W''(z) \right]$
 Which, we can expand $W(z) = \sum_{-\infty}^{\infty} a_n z^n$
 $W'(z) = \sum_{-\infty}^{\infty} n a_n z^{n-1}$, $W''(z) = \sum_{-\infty}^{\infty} n(n-1) a_n z^{n-2}$

$\delta S_F = (h+1) \lambda_0 F$

$\delta S_F = \int d^4x \left[(h+1) \lambda_0 F W'(z) + \lambda_0 F \sum_{-\infty}^{\infty} n a_n \delta(z^{n-1}) \right]$
 $= \int d^4x \left[(h+1) \lambda_0 F W'(z) + \lambda_0 F \sum_{-\infty}^{\infty} n(n-1) a_n z^{n-2} (\lambda_0 h z) \right]$
 $= \int d^4x \left[(h+1) \lambda_0 F W'(z) + \lambda_0 F \sum_{-\infty}^{\infty} n(n-1) a_n z^{n-1} (\lambda_0 h) \right]$

$$= \int d^4x \sum_{n=0}^{\infty} a_n \lambda_D F z^{n-1} ((n+1) + (n-1)h)$$

$$w \sim a z^3 \Rightarrow \delta w = \lambda_D 3a h z^3$$

$$\Rightarrow h+1 + (n-1)h = 4 \quad \text{since } c: h_c = -4$$

$$h n + 1 = 4 \Rightarrow \boxed{n = 3}$$

$$\boxed{W(z) = c z^3} \quad \text{This has weight } \underline{h_w = 3}$$

$\Rightarrow S_F$ from above must have a chiral field with lowest weight $h_z = 3$. Thus, we must replace z w/ z^3 in S_F .

For $\chi = 0$,

$$W(z): \int d^4x e^{(3c z^2 \text{Re}\{F\})}$$

$\rightarrow$ This is for Global supersymmetry. However, note that

$$\text{Re}\{F\} \mapsto c \text{Re}\{F + \frac{1}{12} \bar{\psi}_\mu \gamma^{\mu\nu} \psi_\nu + \frac{1}{2} z \gamma^{\mu\nu} P_R \psi_\nu\}$$

Super potential term, for (S-Gauge $\chi=0$)

$$\int d^4x e^{3c z^2 \text{Re}\{F + \frac{1}{2} z \gamma^{\mu\nu} P_R \psi_\nu\}}$$

$$\int d^4x e^{(3c z^2 \text{Re}\{F\} + \frac{3}{2} z^3 \gamma^{\mu\nu} \text{Re}\{P_R \psi_\nu\})}$$

$$\text{Re}\{P_R \psi_\nu\} \Rightarrow \frac{1}{2} (1 - \gamma_4) \psi_\nu \mapsto \frac{1}{2} \psi_\nu$$

$$\therefore \boxed{S_c := \int d^4x e \left[3c z^2 \text{Re}\{F\} + \frac{3}{4} c z^3 \bar{\psi}_\mu \gamma^{\mu\nu} \psi_\nu \right]}$$

* The full conformal theory is given by

$$S_{\text{CSL}} = S_{S_G} + S_c$$

With S_c F has slightly more complex dynamics i.e

$$\text{its E.O.M:} \quad \text{Re}(F) = \frac{1}{2} (F + \bar{F})$$

$$S_{Gc} \sim \int d^4x e(-F\bar{F})$$

$$\Rightarrow \underline{\frac{\delta S}{\delta \bar{F}} = -F + \frac{3}{2} c z^2 = 0}$$

$$\therefore F = \frac{3}{2} c z^2 = \frac{9c}{2\kappa^2} \quad (\text{On-shell})$$

Plugging this back in we find

$$S_{SG} \sim -\left(\frac{18}{4} \frac{c^2}{\kappa^4}\right)$$

$$S_c = \int d^4x e \left(3c \left(\frac{3}{\kappa^2} \right) \left(\frac{9c}{2\kappa^2} \right) + \frac{1}{4} c \frac{3^{3/2}}{\kappa^3} \bar{\Psi}_\mu \gamma^{\mu\nu\rho} \Psi_\nu \right)$$

$$\therefore S_{SG} + S_c = \int d^4x e \left(\frac{1}{2\kappa^2} \left[R(W(e, \psi)) - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu(W(e, \psi)) \Psi_\rho \right. \right. \\ \left. \left. - \left(\frac{81}{2} \frac{c^2}{\kappa^2} \right) + \frac{(3^{3/2})c^2}{\kappa^2} + \frac{1}{2} c \frac{3^{3/2}}{\kappa} \bar{\Psi}_\mu \gamma^{\mu\nu} \Psi_\nu \right] \right)$$

$$S_{SG} + S_c = \int d^4x \frac{e}{2\kappa^4} \left[R(W(e, \psi)) - \Lambda \right] - \bar{\Psi}_\mu \gamma^{\mu\nu\rho} D_\nu(W(e, \psi)) \Psi_\rho + m_{3/2} \bar{\Psi}_\mu \gamma^{\mu\nu} \Psi_\nu$$

$$\Lambda := -\frac{81c^2}{2\kappa^2}$$

$$m_{3/2} := -\frac{1}{L} = \frac{1}{2} \kappa^2 c z^3 = \frac{3^{3/2} c}{2\kappa}$$

$$\Lambda = -6m_{3/2}^2$$

→ This is an Anti-de Sitter (AdS₄) super(conformal) gravity with parameter L . Thus the superpotential $W(z)$ parameter c determines the value of the cosmological constant Λ .

$$\Rightarrow \text{For } F = \frac{3}{2} c z^2 = \frac{3c}{2\kappa^2} \text{ i. } P_L \eta = \frac{1}{2} P_L A \epsilon - \frac{\kappa}{2\sqrt{3}} F P_L \epsilon$$

The S-susy contributes to the Q-susy of Ψ_μ

Claim: This leads to $\delta \Psi_\mu = \hat{D}_\mu \epsilon$, where

$$\hat{D}_\mu \epsilon := \left(D_\mu - \frac{1}{2L} \gamma_\mu \right) \epsilon = \left(D_\mu + \frac{1}{4} W_{\mu ab} \gamma^{ab} - \frac{1}{2L} \gamma_\mu \right) \epsilon$$

Proof: We work on shell w/ $A_a = 0$, $\lambda_T = \lambda_0 = 0$

$$\begin{aligned} \delta_Q \Psi_\mu &= \left(D_\mu + \frac{1}{4} W_{\mu}^{ab} \gamma_{ab} \right) \epsilon - \gamma_\mu \left(-\frac{\kappa}{2\sqrt{3}} \left(\frac{9c}{2\kappa^2} \right) \right) \epsilon \\ &= D_\mu \epsilon + \frac{3^{3/2} c}{4\kappa} \gamma_\mu \epsilon = D_\mu \epsilon + \left(\frac{1}{2} \left(-\frac{1}{L} \right) \right) \gamma_\mu \epsilon \\ &= \left(D_\mu + \frac{1}{4} W_{\mu}^{ab} \gamma_{ab} - \frac{1}{2L} \gamma_\mu \right) \epsilon \quad \square \end{aligned}$$

Claim: The supersymmetry algebra of the preserved ϵ susy

Contains a Lorentz transformation w/ parameter

$$\lambda^{ab} = \left[\frac{1}{4L} \right] \bar{G}_2 \gamma^{ab} G_2$$

Max

The AdS_0 super algebra: $N=1$, $\{Q_a, Q_b\} = -\frac{1}{2} (\gamma^{\mu})_{ab} P_\mu - \frac{1}{8L} (\gamma^{\mu})_{ab}$

Proof: $P_L \chi = \frac{1}{2} i P_L A \epsilon - \frac{5}{2\sqrt{3}} F P_L \epsilon$

$$[\delta_s(\chi), \delta_a(\epsilon)] P_L \chi = \delta_s \left(\frac{1}{2} P_L (\not{D} \epsilon + F) \epsilon + \sqrt{2} h \not{Z} P_L \chi \right) - \delta_a (\sqrt{2} h \not{Z} P_L \chi)$$

$$= \delta_s \left(\frac{1}{2} P_L F \epsilon + \sqrt{2} \not{Z} \left(\frac{1}{2} \right) \right) = -\sqrt{2} h \left(\frac{-\kappa}{2\sqrt{3}} \right) \delta_s \rightarrow 0$$

$\rightarrow$ Come back to this.